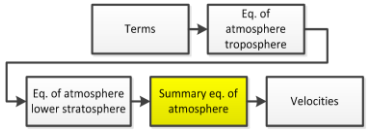
Introduction To Flight Mechanics & Control

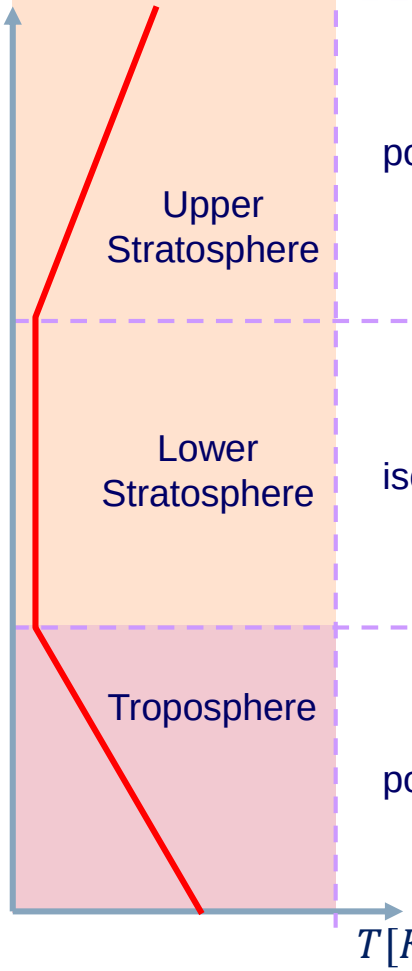
Exercise Session 2



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Summary equations of atmosphere
Standard Atmosphere– ISO 2533



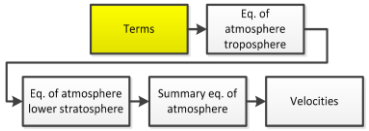


Layer	Features Of Layer	Atmosphere Equations	Data	Height
Upper Stratosphere	polytrope	$\frac{T}{T_{20}} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_{20}}(H_G - H_{G20})\right] = \left[1 + \gamma_{OStr}\frac{(H_G - H_{G20})}{T_{20}}\right]$ $\frac{p}{p_{20}} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_{20}}(H_G - H_{G20})\right]^{\frac{n}{n-1}}$ $\frac{\rho}{\rho_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}(H_G - H_{GR})\right]^{\frac{1}{n-1}}$ $n_{OStr} = 0{,}972$ $\gamma_{OStr} = +1{,}0e-3$	$T_{20} = 216{,}65K$ $\rho_{20} = 0{,}088\frac{kg}{m^3}$ $p_{20} = 547{,}488Pa$ $H_{G20} = 20000m$	20 to 32 km
Lower Stratosphere	isotherm	$\frac{T}{T_{11}} = 1$ $\frac{p}{p_{11}} = e^{-\frac{g_S}{R\,T_{11}}(H_G - H_{G11})}$ $\frac{\rho}{\rho_{11}} = e^{-\frac{g_S}{R\,T_{11}}(H_G - H_{G11})}$ $n_{UStr} = 1$ $\gamma_{UStr} = 0$	$T_{11} = 216{,}65K$ $\rho_{11} = 0{,}364\frac{kg}{m^3}$ $p_{11} = 22632Pa$ $H_{G11} = 20000m$	11 to 20 km
Troposphere	polytrope	$\frac{T}{T_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}H_G\right] = \left[1 + \frac{\gamma_{Tr}\,H_G}{T_S}\right]$ $\frac{p}{p_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}H_G\right]^{\frac{n}{n-1}}$ $\frac{\rho}{\rho_S} = \left[1 - \frac{n-1}{n}\frac{g_S}{R\,T_S}H_G\right]^{\frac{1}{n-1}}$ $n_{Tr} = 1{,}235$ $\gamma_{Tr} = -6{,}5e-3$	$T_S = 288{,}15K$ $\rho_S = 1{,}225\frac{kg}{m^3}$ $p_S = 101325Pa$ $H_{GS} = 0m$	-2 to 11 km

$g_S = 9{,}80665\frac{m}{s^2}$ $\quad R = 287{,}05\frac{J}{kg\,K}$

$$n = \frac{1}{1 + \frac{\gamma\,R}{g_S}} = \frac{g_S}{g_S + \gamma\,R}$$

$$\gamma = \frac{dT}{dH_G} = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$



- Geometric Height h

  Height over WGS-84 reference ellipsoid (“height over sea level”)

- Absolute Height h_a

  Distance to the center of earth (important in astronautics)

- Pressure Height $h = h(p)$

  The height in the standard atmosphere, where the pressure equals the given pressure

  ⇨ *Important for altimeters*

- Temperature Height $h = h(T)$

  Height in standard atmosphere, in which temperature equals given temperature

  ⇨ *Important for calculation Mach number*

- Density Height $h = h(\rho)$

  Height in standard atmosphere, in which density equals given density

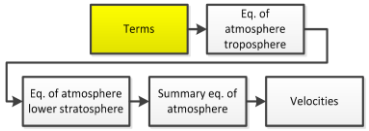
  ⇨ *Important for calculating performance and flight control*

- Geopotential Height H_G

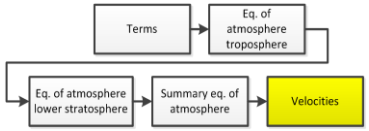
  Defined as the height that takes changes of gravity with height into account

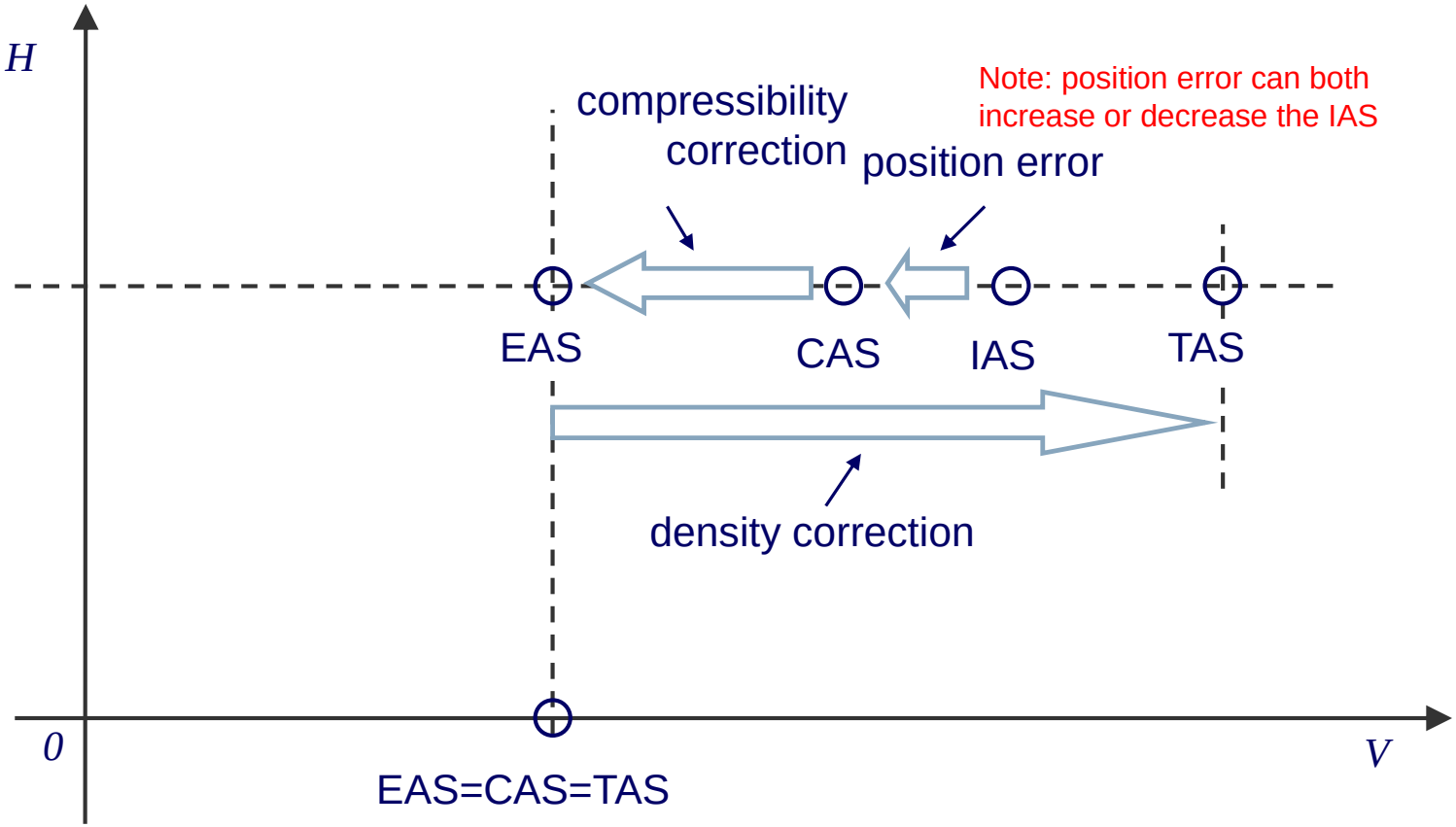
  ⇨ *Further definition follows below*

Airspeeds



- Indicated Airspeed (IAS):
  - Flight speed that is **measured in the airplane**
  - Used in flight operations (stall, max. speed, etc.)
- Calibrated Airspeed (CAS):
  - CAS **corrects instrument errors** in IAS
  - Errors that occur because of a wrong placement of the „Static Port“ (measuring static pressure) are compensated
- Equivalent Airspeed (EAS):
  - EAS **corrects compressibility effects**
  - EAS doesn‘t change with height when flying with a constant dynamic pressure, used in R&D (flight envelopes)
- True Airspeed (TAS):
  - EAS with additional **compensation of density variation with altitude**
  - **Used for performance calculations** (curve radii)


Terms
Eq. of atmosphere troposphere
Eq. of atmosphere lower stratosphere
Summary eq. of atmosphere
Velocities


H
compressibility correction
position error
Note: position error can both increase or decrease the IAS
EAS
CAS
IAS
TAS
density correction
0
V
EAS=CAS=TAS

Key Facts Exercise

The **indicated airspeed** of the aircraft is $V_{IAS} = 320{,}8\,kts$. The atmosphere is assumed to be the **standard atmosphere**. The **density of the surrounding air** is $\tilde{\rho} = 0{,}916\,\frac{kg}{m^3}$.

For this aircraft the following relation holds: $V_{CAS} = 1{,}01 \cdot (V_{IAS} - 2\,kts)$.

Note: We expect that you know these quantities in the exam!

MSL density ISA standard atmosphere: $\rho_{s,MSL} = 1.225\,\frac{kg}{m^3}$

MSL pressure ISA standard atmosphere: $p_{s,MSL} = 1.01325e5\,Pa$

MSL temperature ISA standard atmosphere: $T_{s,MSL} = 288.15\,K$

Gravitational acceleration: $g_s = 9.80665\,\frac{m}{s^2}$

Speed of sound MSL: $a_{s,MSL} = 340\,\frac{m}{s}$

Specific gas constant: $R = 287.05\,\frac{J}{kg\,K}$

Polytropic exponent: $n = 1.235$

$\kappa = 1{,}4$

What is the **geometric height** of the aircraft?

- **Geopotential height H_G**
  - Defined as the height that takes changes of gravity with height into account.

$$H_G = \frac{r_E \cdot h}{r_E + h}$$

Earth radius according to ISO 2533: $r_E = 6356766m$
Geodetic/**geometric height**: h

Thus, given the **geopotential height** we can compute the geometric height.

- Can we compute the geopotential height from the given data?

What is the **geometric height** of the aircraft?

- **Temperature Equations in the troposphere:**

$$T(H_G) = T_{MSL} + \gamma\, H_G$$, where $$\gamma = \frac{dT}{dH_G} = -\left(\frac{n-1}{n}\right)\frac{g_S}{R}$$

or $$n = \frac{1}{1+\frac{\gamma\, R}{g_S}} = \frac{g_S}{g_S + \gamma\, R}$$

We only need the temperature $T(H_G)$ to compute the geopotential height H_G.

- **Basic thermodynamics (Assumption: ideal gas)**

$$\frac{\rho(H_G)}{\rho_R} = \left(\frac{T(H_G)}{T_R}\right)^{-\left[\frac{g_S}{\gamma\, R}+1\right]}$$

$$\frac{\rho(H_G)}{\rho_R} = \left(\frac{T_R + \gamma\, H_G}{T_R}\right)^{-\left[\frac{g_S+\gamma\, R}{\gamma\, R}\right]}$$

$$\frac{\rho(H_G)}{\rho_R} = \left(1 + \frac{\gamma\, H_G}{T_R}\right)^{\frac{1}{n-1}}$$

$$T(H_G) = T_R + \gamma\, H_G$$

$$n - 1 = -\frac{\gamma\, R}{g_S + \gamma\, R}$$

$$H_G = \left[\left(\frac{\rho(H_G)}{\rho_{S,MSL}}\right)^{n-1} - 1\right] \cdot \frac{T_{S,MSL}}{\gamma} = 2927{,}1\text{m}$$

What is the **geometric height** of the aircraft?

- **Finally we compute the geometric height from the geopotential height**

$$H_G = \frac{r_E \cdot h}{r_E + h}$$

$$h = \frac{r_E \cdot H_G}{r_E - H_G} = 2928{,}5m$$

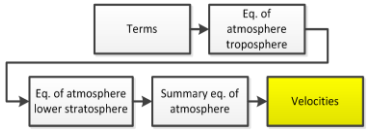
Calculate the **calibrated airspeed**.

For this aircraft the following relation holds: $V_{CAS} = 1{,}01 \cdot (V_{IAS} - 2\ kts)$.

$$V_{CAS} = 1{,}01 \cdot (V_{IAS} - 2kts) = 321{,}99kts = 165{,}64\frac{m}{s}$$

Exercise 2.3:
Can we assume the air to be **incompressible**?

Mach Number and Mach Regions (1)

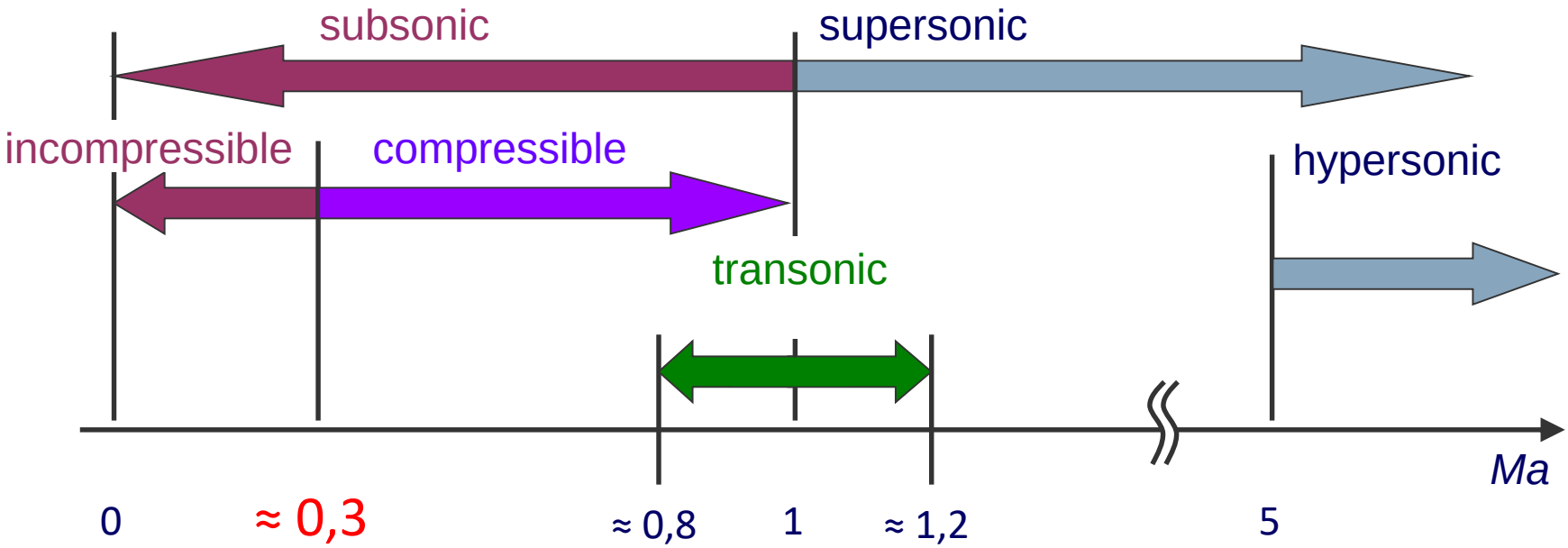


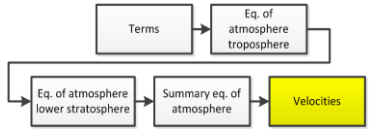
- Definition: Ratio of airspeed to speed of sound

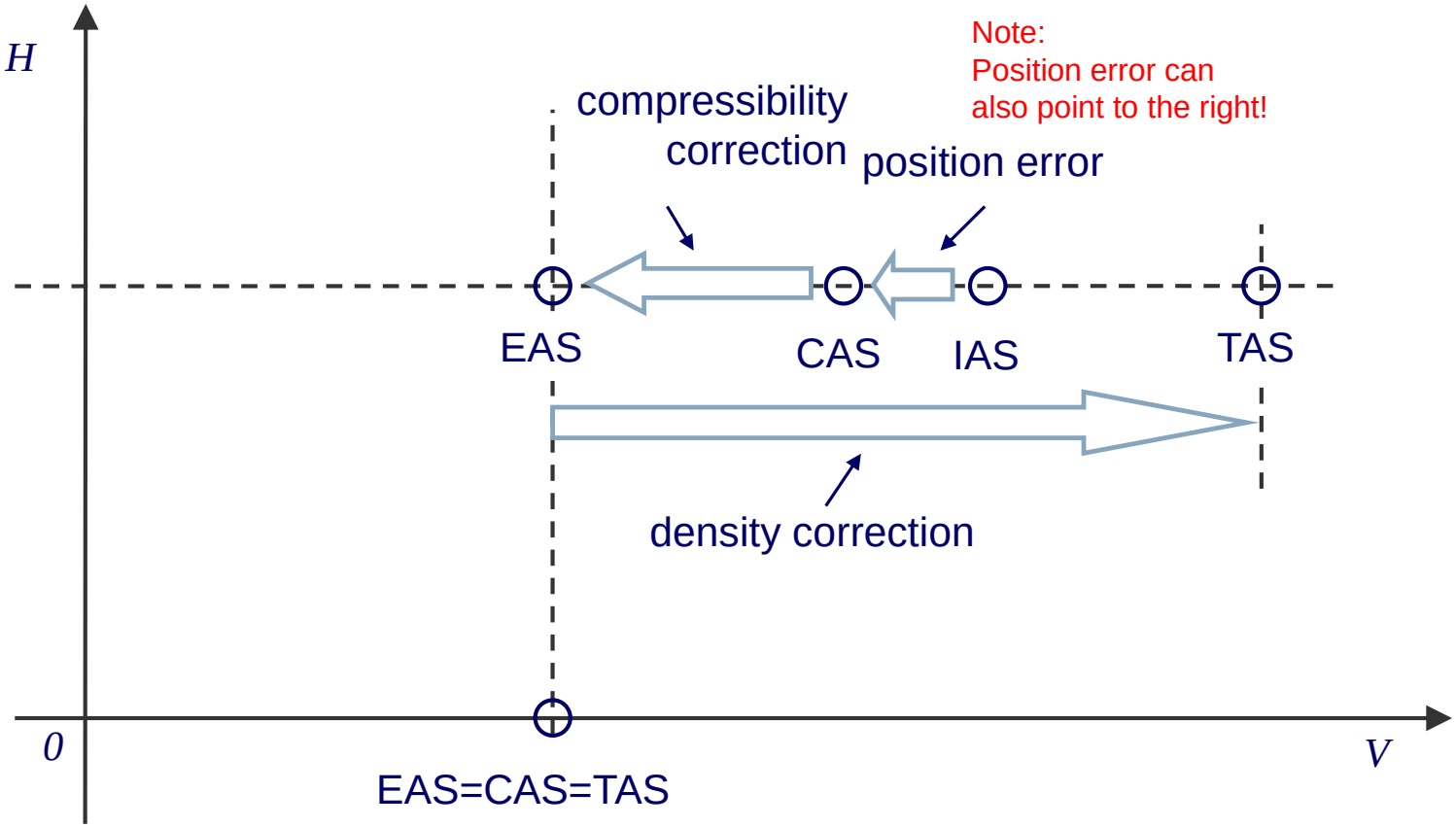
$$Ma = \frac{v}{a}$$

Note:
We use the true airspeed to compute the Mach number!

- Mach Regions:




Terms
Eq. of atmosphere troposphere
Eq. of atmosphere lower stratosphere
Summary eq. of atmosphere
Velocities


H
compressibility correction
position error
Note:
Position error can also point to the right!
EAS
CAS
IAS
TAS
density correction
0
V
EAS=CAS=TAS

Can we assume the air to be **incompressible**?

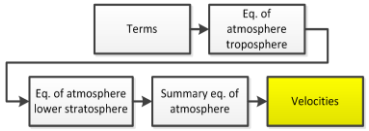
$$Ma = \frac{V_{TAS}}{a} > \frac{V_{CAS}}{a} = 0.5$$

With the speed of sound (at our altitude!!)

$$a = \sqrt{\kappa R\, T(H_G)} = 328.87\ m/s$$

$$T(H_G) = T_{S,MSL} + \gamma H_G = T_{S,MSL} + \left(-0.00650\frac{K}{m} * 2927{,}1m\right)$$

$$= 269{,}12K\ (-4{,}03°C)$$



	Incompressible	Compressible
CAS	$V_{CAS} = \sqrt{\frac{2 \cdot (p_T - p)}{\rho_{MSL}}}$	$V_{CAS} = a_{MSL} \sqrt{\frac{2}{\kappa - 1}\left(\left(\frac{(p_T - p)}{p_{MSL}} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)}$
EAS	$V_{EAS} = V_{CAS}$	$V_{EAS} = a \sqrt{\frac{2}{\kappa - 1} \cdot \frac{\rho}{\rho_{MSL}}\left(\left(\frac{(p_T - p)}{p} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)} = V_{TAS}\sqrt{\frac{\rho}{\rho_{MSL}}}$
TAS	$V_{TAS} = V_{EAS}\sqrt{\frac{\rho_{MSL}}{\rho}}$	$V_{TAS} = a \sqrt{\frac{2}{\kappa - 1}\left(\left(\frac{(p_T - p)}{p} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)} = V_{EAS}\sqrt{\frac{\rho_{MSL}}{\rho}}$

Note: We expect that you know these formulas in the exam!

Calculate the **difference between total and static pressure.**

$$V_{CAS} = a_{MSL}\sqrt{\frac{2}{\kappa-1}\left(\left(\frac{p_T - p}{p_{MSL}}+1\right)^{\frac{\kappa-1}{\kappa}}-1\right)}$$

$$p_T - p = \left[\left(\left(\frac{V_{CAS}}{a_{MSL}}\right)^2 \cdot \frac{\kappa-1}{2}+1\right)^{\frac{\kappa}{\kappa-1}}-1\right]\cdot p_{MSL} = 17857{,}82Pa$$

Calculate the **equivalent airspeed**.

Compressible case!

$$V_{EAS} = \sqrt{\frac{2}{\kappa - 1} \cdot \frac{\kappa \cdot p(H_G)}{\rho_{S,MSL}} \left(\left(\frac{p_T - p}{p(H_G)} + 1 \right)^{\frac{\kappa-1}{\kappa}} - 1 \right)} = 163{,}87 \frac{m}{s}$$

$$p(H_G) = p_{S,MSL} \cdot \left(1 + \frac{\gamma}{T_{S,MSL}} (H_G) \right)^{\frac{n}{n-1}} = 7.0764\text{e}+04\ Pa$$

Relation T - p

Calculate the **true airspeed**.

Compressible case! (same relation holds in the incompressible case)

$$V_{TAS} = V_{EAS}\sqrt{\frac{\rho_{S,MSL}}{\tilde{\rho}}} = 189{,}50\,m/s$$